

Assignment 8 - Threads and mutexes

24.3.2024

Sent

100%

Forsøk 1



Se gjennom tilbakemelding

4.4.2024

Forsøk 1 Poengsum:

100%



Legg til kommentar

Anonym vurdering: **nei**

Ubegrenset antall forsøk tillatt

16.12.2023 til 29.4.2024

▼ Detaljer

Preparation

1. Create a new program for target DISCO-L475VG-IOT01A (B-L475E-IOT01A) under your git folder (`<course folder>/assignments/`) called `assignment_8`. If you create this project based on an example, pick `mbed-os-example-blinky`. If you just copy a previous assignment, make sure the `mbed_app.json` file (if it exists) does not contain the line:
`"requires": ["bare-metal"],`
2. In this assignment no network connectivity is required, so you might consider shrinking compile time by downloading this [.mbedignore](https://uia.instructure.com/courses/14984/files/2350764?wrap=1) (<https://uia.instructure.com/courses/14984/files/2350764?wrap=1>).  (https://uia.instructure.com/courses/14984/files/2350764/download?download_frd=1) file, unzip and add it next to your `main.cpp` file in the project.

Resources

- See the lecture for relevant resources and examples.

Requirements

1. 10% Create a struct with three variables: A minute counter, a second counter and a mutex
2. 5% Create an instance of the struct in the `main()` function
3. 20% Create three threads. Increase the priority of the first two
Pass the struct instance to the three threads as an argument (see the last example in the [documentation](https://os.mbed.com/docs/mbed-os/v6.16/apis/thread.html)  (<https://os.mbed.com/docs/mbed-os/v6.16/apis/thread.html>))
4. 20% Lock the mutex in the three threads so only one thread can access it at a time
5. 15% Increase the minute counter once per minute in the first thread
6. 15% Increase the second counter once per second in the second thread
7. 15% Print the current clock value twice per second in the third thread

Delivery

Deliverer the following as text in Canvas:

- Link to Bitbucket where to code has been delivered.
Directly to the file if there is only 1, or the directory if more than 1.
- Link to YouTube (or other video service) with video showing each of the requirements.
Make sure that each requirement you have completed is clearly shown.
You don't need to show the actual code, just which of the requirements are met.